

Cryptography in Cloud Security

Generating a self-signed SSL certificate

1. Using the `ssl` Commands to generate these files:

```
openssl genpkey -algorithm RSA -out localhost.key
```

```
openssl req -new -key localhost.key -out localhost.csr
```

```
openssl x509 -req -in localhost.csr -signkey localhost.key -out localhost.crt
```

[illegible]

You need to fill these information in order to generate the SSL certificate

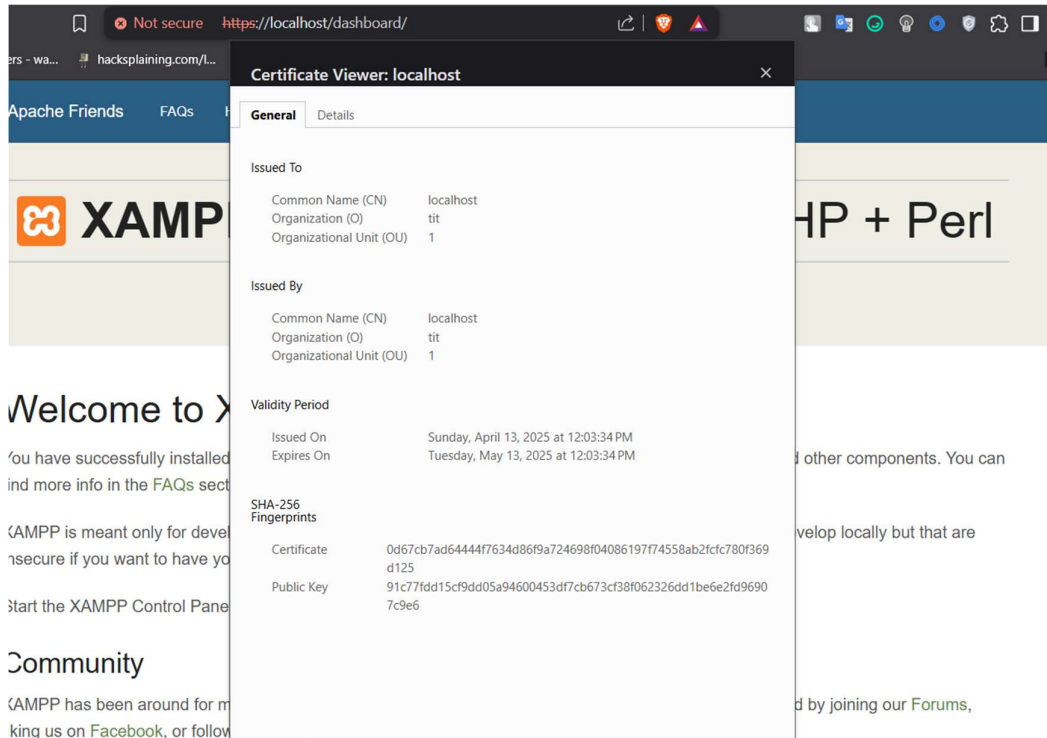
- ## 2. Installing the certificate

To install the certificate you need to go to

C:\xampp\apache\conf

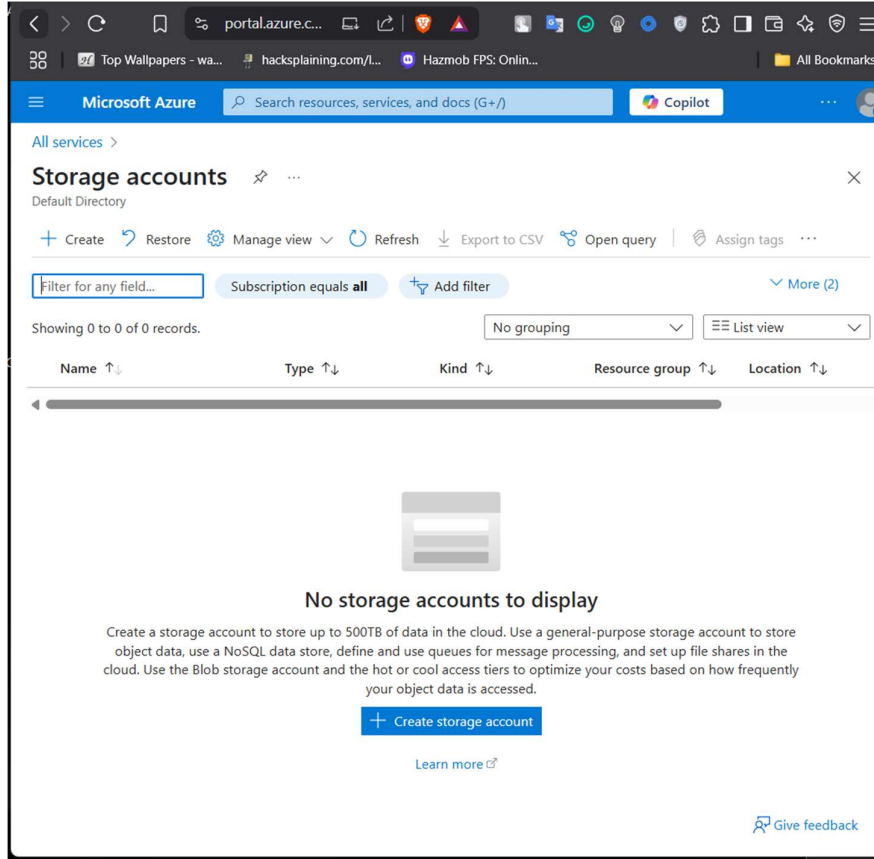
And paste the generatex files from the first commands in the respective folder

After installation it should look like this



Encryption for a cloud storage bucket

1. Search for Storage Accounts



2. Create a Storage Account if you don't have one

The screenshot shows the 'Create a storage account' form in the Microsoft Azure portal. The form is titled 'Create a storage account' and includes a close button. Below the title, there's a section for 'Project details' with instructions to select a subscription and resource group. The 'Subscription' dropdown is set to 'Azure for Students', and the 'Resource group' dropdown is set to '(New) Test'. There's a 'Create new' link below the resource group dropdown. The 'Instance details' section includes fields for 'Storage account name' (set to 'test1'), 'Region' (set to '(Asia Pacific) Southeast Asia'), 'Primary service' (set to 'Select a primary service'), 'Performance' (with 'Standard' selected), and 'Redundancy' (set to 'Geo-redundant storage (GRS)'). The 'Standard' performance option is described as 'Recommended for most scenarios (general-purpose v2 account)', and the 'Premium' option is described as 'Recommended for scenarios that require low latency.' The 'Redundancy' dropdown is set to 'Geo-redundant storage (GRS)'. At the bottom of the form, there's a checkbox labeled 'Make read access to data available in the event of regional unavailability' which is checked. At the very bottom, there are navigation buttons: 'Previous', 'Next', and 'Review + create'. There is also a 'Give feedback' link.

3. Review + Create

The screenshot shows the 'Create a storage account' page in the Microsoft Azure portal. The 'Review + create' tab is selected. The page displays configuration details for a new storage account named 'testresource1test' in the 'Southeast Asia' region. The 'Basics' section includes fields for Subscription (Azure for Students), Resource group (Test), Location (Southeast Asia), Storage account name (testresource1test), Primary service, Performance (Standard), and Replication (Read-access geo-redundant storage (RA-GRS)). The 'Advanced' section shows settings for hierarchical namespace, SFTP, network file system, cross-tenant replication, access tier (Hot), and large file shares (Enabled). Navigation buttons for 'Previous', 'Next', and 'Create' are at the bottom, along with a 'Give feedback' link.

Basics	
Subscription	Azure for Students
Resource group	Test
Location	Southeast Asia
Storage account name	testresource1test
Primary service	
Performance	Standard
Replication	Read-access geo-redundant storage (RA-GRS)

Advanced	
Enable hierarchical namespace	Disabled
Enable SFTP	Disabled
Enable network file system v3	Disabled
Allow cross-tenant replication	Disabled
Access tier	Hot
Enable large file shares	Enabled

4. Check for Encryption in Encryption panel

The screenshot shows the 'Encryption' panel for a storage account named 'tessract' in the Microsoft Azure portal. The 'Encryption' tab is selected. The panel displays information about storage service encryption, including a note that new data will be encrypted and existing files will be encrypted retroactively. The 'Encryption selection' section shows that 'Enable support for customer-managed keys' is set to 'Blobs and files only' and 'Infrastructure encryption' is 'Enabled'. The 'Encryption type' section shows that 'Microsoft-managed keys' is selected. A 'Save' button is at the bottom.

Encryption Encryption scopes

Storage service encryption protects your data at rest. Azure Storage encrypts your data as it's written in our datacenters, and automatically decrypts it for you as you access it.

Please note that after enabling Storage Service Encryption, only new data will be encrypted, and any existing files in this storage account will retroactively get encrypted by a background encryption process. [Learn more about Azure Storage encryption](#)

Encryption selection

Enable support for customer-managed keys	Blobs and files only
Infrastructure encryption	Enabled
Encryption type	☒ Microsoft-managed keys ☐ Customer-managed keys